Reshuffled, Not Ruptured: Brexit, Cross-Border Syndication, and the Identification of Mechanisms under Market-Wide Shocks

Stephen Dossou

Mines Paris – PSL

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Abstract

Did Brexit break the co-investment ties between British and continental European venture capitalists? Using Crunchbase data and an exposure-based event study with wild-cluster bootstrap inference, we find one effect and four non-effects, and we argue the non-effects are the finding. Continental VCs with high pre-determined exposure to UK partners durably reduced the share of their syndicates including a British investor, by 8.2 percentage points against a base rate of 18.7% (95% CI $[-13.1, -3.2]$); among thirteen partner-country corridors subjected to an identical, pre-specified protocol, the UK is the only one to display this pattern. Nothing else moved. Established UK–continental dyads were no more likely to dissolve than comparable continental dyads (triple difference: $+0.3$ pp, $p = 0.77$); the aggregate volume of jointly syndicated UK–continental deals tracked control corridors exactly (95% CI $[-20\%, +26\%]$); and the activity response of exposed VCs fails the same falsification battery that certifies the first stage, appearing with equal or greater strength for exposure to US, Swiss and non-EU partners, and materialising only after 2020. Nor is there a mirror image: run on a British universe, the same battery leaves the continental aggregate unidentified (pre-trends $p = 0.04$) while British withdrawal from *American* partners clears the grid outright (-5.0 pp, pre-trends $p = 0.95$), so the symmetry an earlier draft of this paper reported does not survive. The composition of European syndicates was reshuffled; no relationship was ruptured. The broader lesson is methodological: when a shock treats every agent on the same date, designs without an untreated unit do not merely lose precision—they manufacture mechanisms.

JEL codes: G24, F36, D85, G15.
Keywords: Venture capital, Brexit, cross-border investment, syndication, identification, null results.

1 Introduction

Venture capital deals are syndicated, and syndication is relational. A co-investment tie is not a spot transaction but an asset: its value is the flow of deals the partnership is expected to generate. Before June 2016, British venture capitalists were the most sought-after partners in Europe, combining reputation, capital, and access to the deepest European market. The referendum of 23 June 2016 was substantially unanticipated by financial markets (Breinlich et al., 2018); sterling fell eight percent overnight and UK policy uncertainty reached a record (Baker et al., 2016). It is natural to expect that the ties binding British and continental investors were among the casualties.

They were not. This paper documents one effect and four non-effects, and argues that the non-effects are the substance.

The effect is compositional. Among continental VCs that all had at least one British co-investor before the vote, those whose pre-determined reliance on UK partners was above the median durably reduced the share of their syndicates including a British investor—by 8.2 percentage points ($p < 0.001$; 95% CI $[-13.1, -3.2]$) against a control base rate of 18.7%, a relative decline of forty-four percent. Both joint pre-trend tests pass, and the effect is negative and significant in each of three post-referendum windows, including 2022–2025. What makes us willing to call it identified is not the pre-trend test, which we show in Section 6 is weaker evidence than it is usually taken to be, but the placebos: submitted to an identical, pre-specified protocol, twelve other partner corridors—the United States, Canada, Israel, Japan, Korea, France, Germany, Switzerland, the Netherlands, Sweden, Belgium, and a non-EU OECD aggregate— produce nothing comparable. The United Kingdom is the only corridor of thirteen that clears the grid.

The non-effects are what that compositional shift did *not* do.

First, it did not break relationships. We construct the universe of investor dyads established before 2014 and ask whether a persistent UK–continental pair is more likely to dissolve after the referendum than a persistent continental–continental pair. The triple difference is $+0.3$ percentage points ($p = 0.77$; 95% CI $[-1.8, +2.5]$ on a semi-annual reactivation base rate of 5.6%). Persistent ties decay everywhere—by 2.3 points on average—and they decay at the same rate whether the partner is in London or in Milan. A naive design that classifies dyads on the full pre-referendum window instead yields -7.1 points, highly significant; but the same design yields -5.1 for France and -8.0 for Germany. That is regression to the mean, not Brexit.

Second, it did not shrink the corridor. We build a panel of country-pair corridors and compare the volume of jointly syndicated deals in the eighteen UK–continental corridors against fifty-five continental–continental corridors. The post-referendum effect is $+0.03$ log points ($p = 0.81$), with a 95% confidence interval that bounds any effect within $[-20\%, +26\%]$. Pre-trends are flat and a 2014 placebo passes.

Third, the activity response of exposed VCs does not survive falsification. The apparent finding—that small, highly exposed continental VCs became *more* active after the vote—is not robust. Passed through the same thirteen-corridor battery that certifies the first stage, it fails twice over. It is generic: exposure to US partners produces +7.4 points, exposure to Swiss partners +5.6, exposure to a non-EU OECD aggregate +7.7, and all three clear the pre-specified grid that the UK does not. And it is mis-timed: in the window that follows the referendum directly (H2 2016 to 2019), the UK coefficient is +3.2 points with $p = 0.18$. The effect materialises only in 2020–21 and 2022–25, during the global venture boom. We show the source: the conventional control group—VCs never exposed to the partner country—is 72% composed of purely domestic, unconnected investors, so the comparison contrasts networked firms with isolated ones, and networked firms capture booms better. Replacing it with an equally networked control halves the estimate and removes its significance.

Fourth, the shift has no mirror image. The natural claim—and the one an earlier version of this paper made—is that a severed corridor is felt at both ends, and a British panel duly returns a first stage of −7.1 points. Run the battery on a British universe, where the prediction is sharper still because Brexit changed the United Kingdom's relationship with the Union and not with anyone else, and the claim inverts. The continental aggregate reproduces in magnitude (−6.0 points, $p = 0.039$) but fails both pre-trend tests ($p = 0.04$, $p = 0.03$): the exposed group was already converging before the vote. The corridors that clear the grid cleanly are the ones Brexit cannot reach—British withdrawal from American partners is −5.0 points with pre-trends at $p = 0.95$, and from Swiss partners −10.9 points—while France, Germany and the Netherlands are null. The earlier draft found symmetry because it classified exposure on the pre-referendum window and then tested pre-trends inside that same window, where the gap between groups is stable by construction. Sections 5 and 7.4 give the diagnosis.

This yields the paper's methodological claim, which we take to be its main contribution. Brexit is a market-wide shock: it hits every European VC on the same date, so there is no untreated unit in the conventional sense. Identification must therefore come from *differential exposure*, and the aggregate must be absorbed by time fixed effects. When it is not—when a design compares pre to post without a contemporaneous counterfactual—the aggregate trend does not simply inflate standard errors. It loads onto the treatment dummy and produces a coefficient that looks like a mechanism, has the right sign, and survives ordinary robustness checks. We give three worked examples of this failure: the dyadic divorce, the activity expansion, and the symmetry of the rupture. All three are artifacts. Only a design with a contemporaneous control detects that they are.

To arrive at these conclusions, we build four panels from Crunchbase, each designed so that treated and control units are subject to the same shock and differ only in predetermined exposure. The first is an investor panel: 448 continental VCs that all had at least one British co-investor before 2014 (216 above the median exposure share, 232 below), observed over 23 semesters from 2014 to 2025. Because a single British tie mechanically pushes a small investor's exposure share above the median, every specification carries a small $\times$ post control that nets out the resulting size trend. The second is a dyad panel: 6,693 investor pairs established before 2014—1,347 with a British partner and 7,224 purely continental— observed over 197,133 dyad-semesters, on which we estimate a triple difference. The third is a corridor panel: 73 country-pair corridors, of which 18 link the United Kingdom to the continent and 55 link two continental countries, with the outcome defined identically for both. The fourth mirrors the first on the other side of the Channel: 255 British VCs that all had at least one continental co-investor before 2014, which lets us ask of a British investor exactly what we ask of a continental one. Exposure is classified using only deals concluded before 2014, so that the classification cannot be contaminated by anticipation, and the semesters 2014–2015 are held out to test whether the groups were parallel before the shock. Inference is by wild-cluster bootstrap-t throughout, with between twenty-one and seventy-three clusters depending on the panel, and confidence intervals are obtained by inverting the bootstrap distribution.

Our contribution is threefold. First, we provide a set of null results—two of them precisely estimated—on a question where the policy narrative has run far ahead of the evidence, and we are explicit about which of our nulls are informative and which merely underpowered. Second, we identify one real effect, a durable compositional reallocation away from British partners, and establish that it is unique to the United Kingdom among thirteen partner corridors subjected to an identical protocol. Third, and we regard this as the most portable contribution, we demonstrate that control-free designs applied to market-wide shocks do not merely lose precision: they manufacture mechanisms. We give three worked examples of the failure, all of them large, all of them significant, and all of them spurious.

Table 1 collects the five results, with the verdict each receives and the reason for it. It is the paper in one page.

Table 1: The five results.

	Outcome	Estimate	95% CI	Verdict
§6	UK-partner rate (pp)	−8.20	[−13.11, −3.15]	**Identified**
§7.1	Dyad survival, triple diff. (pp)	+0.30	[−1.84, +2.51]	Precise null
§7.2	Corridor volume (log pts)	+0.03	[−0.20, +0.26]	Precise null
§7.3	Activity, window [0, 6] (pp)	+3.18	[−1.47, +8.16]	Not identified
§7.4	British mirror, first stage (pp)	−6.01	[−11.52, −0.33]	Not identified

Notes. Wild-cluster bootstrap-t intervals throughout (Appendix A reports base rates and minimum detectable effects). The first stage is identified because, among thirteen partner corridors submitted to an identical pre-specified protocol, the United Kingdom is the only one to clear the grid. The two precise nulls exclude economically meaningful effects: no reduction in the survival of British ties beyond a third of the baseline rate, and no change in corridor volume outside [−20%, +26%]. The two unidentified results are the paper's cautionary half: the activity expansion because it appears with equal strength for American, Swiss and non-EU OECD exposure and only after 2020, the British mirror because both pre-trend tests fail while withdrawal from American partners clears the grid. "Not identified" is not "zero"—the last row's interval excludes zero—and the distinction is the subject of Section 8.

Section 2 situates the paper. Section 3 describes the setting and explains why the AIFMD passport, frequently invoked, cannot be the operative mechanism. Section 4 presents the data. Section 5 sets out the design and the pre-specified falsification protocol. Section 6 presents the first stage. Section 7 presents the three non-effects. Section 8 draws the methodological lesson.

2 Related literature

2.1 Syndication as relational capital

Venture capital deals are overwhelmingly syndicated, and the literature has established why. Lerner (1994) first documented that syndication is not merely risk-sharing: experienced investors syndicate with one another at the first round, which is consistent with a screening motive. Brander et al. (2002) formalise the tension between the two classical rationales—improved venture selection through a second opinion, versus the value added by complementary skills—and find in favour of the latter. Casamatta and Haritchabalet (2007) extend this by showing that syndication is not costless to the junior partner: sharing the equity upside dilutes the marginal return to effort, so an inexperienced VC syndicating with a senior one may exert less effort than it otherwise would. Manigart et al. (2006) document the determinants of the syndication decision across European markets.

What makes syndication ties an object of study in their own right is their persistence. Sorenson and Stuart (2001) show that syndication networks propagate opportunities across geographic space, so that an investor's position in the network determines what it gets to see. Hochberg et al. (2007) demonstrate that better-networked venture capitalists enjoy significantly better fund performance, and Hochberg et al. (2010) that incumbent networks constitute a barrier to entry: a tie is a durable, rivalrous asset. The value of a partnership is therefore not the deal in front of it but the flow of deals it is expected to generate—which is precisely why an uncertainty shock to the future of a partnership should, in principle, destroy value today, and why the absence of any such destruction in our data is worth establishing carefully.

A parallel strand studies what investors bring to the companies they back, and thereby what a portfolio company loses when an investor exits a syndicate. Hellmann and Puri (2002) show that venture capitalists professionalise their portfolio companies; Gompers (1995) that they monitor through staging; Bernstein et al. (2016) that this monitoring is causal, using an instrument based on travel times. Reputation carries value that transfers to the company: Megginson and Weiss (1991) and Stuart et al. (1999) establish the certification effect at IPO, Hsu (2004) shows that entrepreneurs pay for affiliation with high-reputation investors, and Nahata (2008) that investor reputation predicts portfolio performance. If British investors were disproportionately providing certification to continental syndicates—a natural conjecture given London's position—then their withdrawal should have degraded the downstream outcomes of the companies these syndicates backed. Appendix B shows it did not.

2.2 Cross-border venture capital and its frictions

Venture capital is stubbornly local. Cumming and Dai (2010) document a pronounced local bias in investment; Sorenson and Stuart (2001) attribute part of it to the structure of the syndication network itself. Cross-border investment therefore requires a mechanism to overcome distance, and syndication with a local partner is the canonical one: Tykvová and Schertler (2012) show that syndicating with a local VC moderates the liability of foreignness for the cross-border investor. Schertler and Tykvová (2012) examine what attracts cross-border VC inflows and find institutional and informational determinants to dominate mechanical ones.

The frictions are not only informational but cultural and institutional. Guiso et al. (2009) show that bilateral trust—shaped by history, religion, and conflict— predicts bilateral trade and portfolio investment; Bottazzi et al. (2016) demonstrate that the same is true within European venture capital specifically, where trust between the investor's and the entrepreneur's countries predicts investment. This literature furnishes the strongest prior that Brexit should have mattered: if a political rupture degrades bilateral trust, and trust supports cross-border venture investment, the UK–continental corridor was exposed. Our results indicate that whatever happened to trust, it did not pass through into the volume or the survival of co-investment ties.

2.3 Brexit and the financial sector

The macroeconomic and trade consequences of the referendum are well documented. Baker et al. (2016) show that UK policy uncertainty reached an all-time high; Breinlich et al. (2018) extract the market's revision of expectations from the stock-price response and find it concentrated in domestically oriented firms; Dhingra et al. (2017) quantify the trade effects; Steinberg (2019) models the macroeconomic cost of trade policy uncertainty specifically; Born et al. (2019) estimate the output cost using synthetic control methods. What unites these papers is that they all found a way to construct a counterfactual—a control group of firms, a synthetic UK—and this is not incidental.

Within venture capital, the closest paper is Bosio et al. (2025), published in *Research Policy* and co-authored with the research department of the European Investment Fund. Using the European Data Cooperative at the level of functional urban areas, they document that UK hubs cut their cross-border investment immediately after the announcement while continental hubs increased theirs after enforcement, and that the share of UK deals involving a cross-border syndicate rose substantially, which they read as strategic adaptation. Their object is flows and geography at the level of the urban hub; ours is ties and adaptation at the level of the firm, identified off predetermined exposure and a falsification battery. The two are complementary, and where their descriptive statistics touch ours the profiles agree.

2.4 Identification under market-wide shocks, and the value of nulls

The recent econometric literature on difference-in-differences has focused on the biases that arise from heterogeneous treatment effects and staggered adoption (Goodman-Bacon, 2021; de Chaisemartin and D'Haultfœuille, 2020; Callaway and Sant'Anna, 2021; de Chaisemartin and D'Haultfœuille, 2024), surveyed by Roth et al. (2023). That literature presumes the existence of units that are, at least for a while, untreated. Our setting has none: the referendum reaches every European venture capitalist on the same day. The relevant failure mode is therefore not contamination of the comparison group by early-treated units, but the total absence of a comparison group—in which case the aggregate time trend is not absorbed and loads directly onto the treatment indicator. We know of no systematic treatment of this failure mode, and we regard the three worked examples in this paper as its clearest available illustration.

Our inference follows the literature on clustering with a small number of groups. Cameron et al. (2008) introduce the wild-cluster bootstrap; MacKinnon and Webb (2017) characterise its behaviour under heterogeneous cluster sizes, and MacKinnon et al. (2023) provide the current guide to practice. With between twenty-one and thirty country clusters, asymptotic cluster-robust inference would substantially overreject, and we use the bootstrap—including for confidence intervals, obtained by inverting the bootstrap distribution—throughout.

Finally, this is a paper whose principal contribution is a set of null results, which raises an obvious question about why such papers are rare. Brodeur et al. (2016) document the discontinuity in the distribution of published test statistics around conventional significance thresholds; Andrews and Kasy (2019) formalise the resulting selection and show how severely it distorts the published record. A null is publishable when it is precise and when it speaks to a claim that has been made. We have attempted to satisfy both conditions: Appendix A reports what each of our nulls excludes, and the policy narrative that has run ahead of the evidence supplies the claims they bear on. Where a null is *not* precise—as ours on activity is not—we say so rather than presenting an underpowered zero as evidence of absence.

3 Setting

On 23 June 2016, 52% of UK voters chose to leave the European Union. The outcome was not priced: sterling fell eight percent overnight and the UK Economic Policy Uncertainty Index reached an all-time high (Baker et al., 2016; Breinlich et al., 2018). Legal separation followed much later—withdrawal on 31 January 2020, end of the transition period on 31 December 2020—so the four and a half years that concern us most are years of unresolved institutional uncertainty rather than of changed law. Any effect dated to 2016 must therefore run through expectations about the future of the relationship, not through any binding constraint in force at the time.[1]

The shock is therefore composite—regulatory, macroeconomic, and political at once—and we make no attempt to decompose it. Our object is the effect of the referendum on cross-border syndication, not the relative weight of the channels through which it travelled, which our data cannot identify. We estimate what happened; we do not assert a mechanism we cannot test.

4 Data

We use Crunchbase, from which we build an investment-level panel linking each funding round to each participating investor, with the country of the investor's headquarters and of the portfolio company. We restrict rounds to venture types (pre-seed

[1]The constraint most often invoked is the loss of the passport conferred by the Alternative Investment Fund Managers Directive. Whatever its consequences for fund managers, it cannot be the operative channel here: the directive governs the management and marketing of funds to professional investors, not the acquisition of an equity stake in a portfolio company, and it remained in force for the United Kingdom until the end of 2020.

through Series J, angel, and unspecified venture) and to OECD-domiciled targets. The resulting panel contains 1,211,489 investor–round observations.

The unit of time is the semester. We write $\text{sem} = 2 \times \text{year} + \mathbb{1}\{\text{month} > 6\}$ and define τ, event time, relative to 2016H2, the first semester wholly after the referendum. Thus $\tau = 0$ is 2016H2, $\tau = -1$ is the referendum semester itself (which we never use as reference), and $\tau = -2$ is 2015H2, our omitted category.

"Continental Europe" (CE) comprises the European countries excluding the United Kingdom and Ireland. A round is UK-exposed if at least one participating investor is headquartered in the United Kingdom.

Ireland is the only country we remove by name, so it owes the reader a sentence. Irish investors do not stand towards British ones as the rest of the continent does: the land border and the Common Travel Area survived the referendum intact, and Dublin was itself a destination for financial entities leaving London after it, so Irish exposure to British partners can move for reasons that have no continental counterpart. That reason was fixed before we looked, which makes it exactly the kind of reason this paper distrusts elsewhere. We therefore test it rather than rest on it. Adding Ireland brings the universe from 448 investors to 470 and moves the first stage from -8.20 points to -8.01 ($p < 0.001$; 95% CI $[-12.62, -3.26]$), with both pre-trend tests still passing, though less comfortably than before ($p = 0.11$ and $p = 0.18$, against 0.23 and 0.36). A reader who prefers the wider definition may read -8.0 for -8.2 and nothing else follows. Ireland cannot be a corridor in its own right: twenty-two eligible investors, eleven either side of the median, return -6.8 points with an interval of $[-49.8, +29.7]$—eighty points wide—and fall far below the minimum cell that Section 5 requires of every corridor we report. We have run this check on the first stage only. Ireland adds twenty-two investors to four hundred and forty-eight, and we expect the dyad, corridor and British panels to be equally indifferent to it, but we have not re-estimated them and do not claim to have.

Table 2: Summary statistics: the estimation sample.

	High exposure	Low exposure
Investors	216	232
Pre-2014 deals (mean)	10.51	32.48
Pre-2014 deals with a UK partner (mean)	3.59	2.91
UK exposure share (mean)	0.429	0.108
Small (≤ 5 pre-2014 deals), %	58.8	5.2

Notes. Continental VCs with at least two pre-2014 deals and at least one pre-2014 British co-investor. "High" denotes an exposure share above the sample median.

The two groups differ in size: highly exposed investors are, mechanically, smaller, because a given number of British partnerships represents a larger share of a smaller book. We do not regard this as a threat—investor fixed effects absorb the level—

but it does mean that any outcome correlated with size and trending over the sample would contaminate the estimate. We therefore include a $\text{small}_i \times \text{post}_s$ control in every specification, allowing small and large investors to follow different post-referendum trajectories, and we report throughout the pre-trend tests that would detect a failure.

What Crunchbase does not see. Crunchbase is assembled from public announcements, and coverage follows visibility rather than a sampling frame. Three consequences bear on what we can claim. Coverage is thinner at the earliest stages: a pre-seed or seed round that is never announced does not enter, and the omission is not random, since the rounds that go unannounced are those raised by the least visible firms. It is thinner outside the large western markets: a Polish or Portuguese syndicate is less likely to be recorded than a French or German one of the same size. And it records only investors that Crunchbase identifies as organisations with a headquarters country, so unregistered angels and vehicles without a country code enter our denominators without ever triggering a country flag.

None of this is neutral for our estimates, and we would rather state the direction than assert robustness. Our sample is continental VCs with at least two pre-2014 deals and a pre-2014 British partner: firms visible enough to be recorded at least three times. The first stage therefore describes the behaviour of established, observable European investors, and we make no claim about the invisible tail—about angels, about firms whose entire book is unannounced seed rounds, or about the peripheral markets where a single missing round moves an exposure share materially. If Brexit's effect on syndication were concentrated among small, informal, or peripheral investors, our design would not find it. We report a null on the corridor and on the dyads; those nulls are nulls for the venture capital that Crunchbase can see.

Comparison with the European Data Cooperative. The nearest alternative source is the European Data Cooperative, the proprietary database assembled by Invest Europe with national venture capital associations and used by Bosio et al. (2025), which covers European investors only. Crunchbase covers the world, and UK rounds are heavily co-invested by American funds; Crunchbase also records angels and organisations without a country code, which enter our denominators without ever triggering a country flag. Levels therefore differ across the two sources. Our identification does not turn on matching them: it comes from differential exposure within the Crunchbase universe, not from the absolute level of cross-border activity.

5 Design

5.1 No untreated unit

Brexit hits every European venture capitalist on the same date. There is no group that the referendum did not reach. A conventional difference-in-differences is therefore unavailable, and a simple pre/post comparison is uninterpretable: it cannot distinguish the referendum from anything else that happened to European venture capital after June 2016—and a great deal did, culminating in the global venture boom of 2020–2021.

Our identification comes from *differential exposure*. Among agents all treated by the same shock, those whose pre-determined reliance on the affected channel was higher should respond more. Semester fixed effects absorb the common component; the interaction of exposure with event time identifies the relative response. The estimand is a *relative* effect, and we are explicit that it is: we cannot recover the level effect of Brexit on European venture capital, and no design on these data can.

5.2 Specification

For investor i in semester s,

$$y_{is} \;=\; \sum_{\tau \neq -2} \beta_\tau \cdot \mathbb{1}\{s = \tau\} \cdot \text{High}_i \;+\; \alpha_i \;+\; \delta_s \;+\; \varepsilon_{is}, \tag{1}$$

with investor and semester fixed effects. High_i is a pre-determined exposure indicator. Standard errors are clustered on the investor's country, of which there are between twenty-one and thirty depending on the sample—few enough that asymptotic cluster-robust inference is unreliable. We therefore report wild-cluster bootstrap-t p-values throughout (Rademacher weights, unrestricted, equal-tailed, $B = 999$), following Cameron et al. (2008) and MacKinnon and Webb (2017), and we invert the bootstrap distribution to obtain confidence intervals.

5.3 Early classification

Classifying exposure on the full pre-referendum window invites regression to the mean: an investor observed with an unusually high share of UK deals in 2015 will mechanically revert. We therefore classify exposure using *only* deals concluded before 1 January 2014—well before any Brexit signal—and begin the measurement panel in 2014. This leaves $\tau \in \{-5, -4, -3\}$ as a held-out window, used exclusively to test whether the groups were parallel before the shock arrived. Section 7 shows this correction is not cosmetic: it is the difference between a spurious -7.1 point "divorce" and a true zero.

5.4 The falsification protocol

We fix, before looking at any result, a single mechanical protocol applied identically to the United Kingdom and to twelve placebo partner corridors. For each corridor we compute: (i) the static exposure-by-post coefficient; (ii) a joint pre-trend test on $\tau \in \{-5, -4, -3\}$; (iii) a stricter joint pre-trend test additionally including $\tau = -1$, the referendum semester itself; and (iv) three post-referendum windows— $\tau \in [0, 6]$ (H2 2016 to 2019), $\tau \in [7, 11]$ (2020–21), and $\tau \in [12, 17]$ (2022–25).

The reading grid, also fixed ex ante: a genuine Brexit response requires a significant static coefficient, clean pre-trends on *both* tests, and an effect of consistent sign across all three windows including the last. An effect confined to 2020–21 is a pandemic-and-boom phenomenon, not a referendum effect. An effect absent from the first window is not a referendum effect either. The United Kingdom is subjected to exactly the same grid as the placebos, and we report the verdict the grid returns, whatever it is.

6 The first stage: a durable compositional shift

The universe is continental VCs with at least two pre-2014 deals and at least one pre-2014 British co-investor. Every investor in this sample is UK-exposed; the contrast is between those above and below the median exposure share. The outcome huk_{is} equals one if investor i syndicated with a British investor in semester s.

Figure 1 plots Equation (1). The decline begins in the semester containing the referendum and does not reverse over the following nine years.

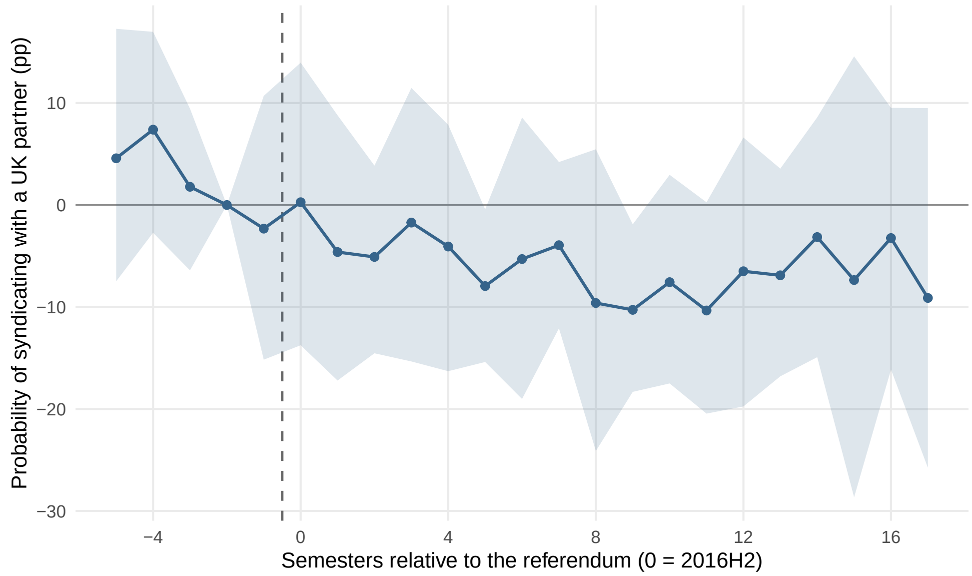

Figure 1: First stage. Probability that a continental VC syndicates with a British partner in a given semester, high versus low pre-2014 exposure. Early classification, investor and semester fixed effects, wild-cluster bootstrap-t bands ($B = 999$, clustered on investor country). Reference semester: 2015H2.

Two features of the figure deserve comment, and we prefer to raise them ourselves. The semester-by-semester coefficients are imprecise: only two of the eighteen post-referendum estimates have a bootstrap interval excluding zero. This is a power problem, not an identification problem—each semester is a single draw from 448 investors—and the effect is identified off the post-period average, which is precisely estimated. Readers who find event-study plots more persuasive than pooled coefficients should be aware that in a panel this size the plot cannot carry the inference.

More seriously, the held-out coefficients are not flat. Relative to 2015H2, the high-exposure group sits 4.6 and 7.4 points *above* the low-exposure group in 2014, and the gap closes monotonically thereafter. The joint pre-trend tests pass—$p = 0.23$ on $\tau \in \{-5, -4, -3\}$ and $p = 0.36$ when the referendum semester is added—but as Roth et al. (2023) emphasises, such tests have low power, and passing one is weaker evidence than the phrase "parallel pre-trends" usually conveys. We therefore state the limitation plainly: we cannot exclude that part of the post-referendum decline continues a downward drift already under way. What argues against that reading is the placebo evidence rather than the pre-trend test—the same specification applied to twelve other partner corridors recovers nothing comparable, and a mean-reverting drift in exposure shares would not single out the United Kingdom.

The pooled result is in Table 3. Highly exposed continental VCs reduce their UK-partner rate by 8.2 percentage points relative to less exposed peers ($p < 0.001$; 95% CI $[-13.1, -3.2]$), against a control base rate of 18.7%: a relative decline of forty-four percent. Both pre-trend tests pass comfortably ($p = 0.23$ and $p = 0.36$). The effect is present in all three windows—−6.4 points in H2 2016–2019, −10.6 in 2020–21, −8.3 in 2022–25—and is therefore durable, not a pandemic artifact.

Table 3: The thirteen-corridor battery, first stage. Identical protocol, fixed ex ante.

Partner	Static	Pre-trends		Windows		Verdict
	(pp)	(i)	(ii)	$[0,6]$	$[12,17]$	
GBR	$\mathbf{-8.2}^{***}$	0.23	0.36	-6.4^{**}	-8.3^{**}	**Durable rupture**
USA	−2.1	0.21	0.16	−0.3	−2.7	null
CAN	+0.5	0.00	0.00	−2.1	+4.2	null
ISR	−0.7	0.27	0.00	+0.8	−1.3	null
JPN	−0.8	0.53	0.11	+0.3	+1.8	null
KOR	−4.6	0.10	0.00	−3.0	−4.0	null
FRA	−3.0	0.43	0.52	−1.3	−5.0	null
DEU	-3.0^{**}	0.47	0.20	+0.4	-5.5^{***}	mixed profile
CHE	-9.2^{***}	0.91	0.05	-10.9^{***}	-9.2^{***}	pre-vote movement
NLD	−0.8	0.10	0.12	+1.5	−0.1	null
SWE	−3.2	0.00	0.00	−3.2	−2.8	null
BEL	-5.2^{**}	0.00	0.00	−1.3	-6.6^{*}	pre-vote movement
Non-EU OECD	−1.5	0.09	0.15	−0.7	−1.5	null

Notes. Outcome: indicator that the investor syndicated with a partner from the corridor country in the semester. Early classification (pre-2014), panel 2014–2025, investor and semester fixed effects, clustered on investor country, wild-cluster bootstrap-t with $B = 999$. Pre-trend (i): joint test on $\tau \in \{-5, -4, -3\}$; (ii): joint test additionally including $\tau = -1$. Verdicts are assigned mechanically by the ex-ante grid of Section 5. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Of thirteen corridors, the UK is the only one to clear the grid. Switzerland and Belgium produce significant coefficients but fail the strict pre-trend test—their investors were already moving before the vote, so nothing is identified. Germany is significant but its profile is mixed. The remaining nine are null. The compositional reallocation away from British partners is real, it is durable, and it is specific to the United Kingdom.

7 What did not happen

The compositional shift is the whole of the effect. This section establishes that it severed no relationship, destroyed no volume, generated no measurable adaptation, and had no counterpart on the British side of the corridor.

7.1 No rupture of established ties

The natural place to look for a divorce is the tie itself. We construct every dyad (i, j) of European investors that co-invested at least once before 2014, label it *persistent* if it shared at least two deals and *occasional* otherwise, and ask whether it is reactivated—shares at least one further deal—in each subsequent semester. The treated dyads are the 1,347 pairs with a British partner; the control dyads are the 7,224 continental–continental pairs. The estimand is the triple difference: does the post-referendum decay of persistent ties differ between the two corridors?

$$\text{active}_{ds} = \beta_1(\text{persistent}_d \times \text{post}_s) + \beta_2(\text{persistent}_d \times \text{post}_s \times \text{UK}_d) + \beta_3(\text{post}_s \times \text{UK}_d) + \Gamma X_{ds} + \alpha_d + \delta_s + \varepsilon_{ds} \tag{2}$$

β_2 is the Brexit effect net of the natural decay of relationships. It is +0.30 percentage points, with $p = 0.77$ and a 95% confidence interval of $[-1.84, +2.51]$ on a semi-annual reactivation base rate of 5.63% (197,133 observations, 6,693 dyads, 29 clusters). We can therefore exclude any reduction in the survival of British ties exceeding 1.8 points—one third of the baseline rate.

Meanwhile $\beta_1 = -2.26$ points: persistent ties decay, everywhere, at the same rate regardless of the partner's nationality. The event-study version of Equation (2), estimated separately by corridor (Figure 2), shows the two curves lying on top of one another; if anything, in the later semesters the continental control decays somewhat faster than the British corridor.

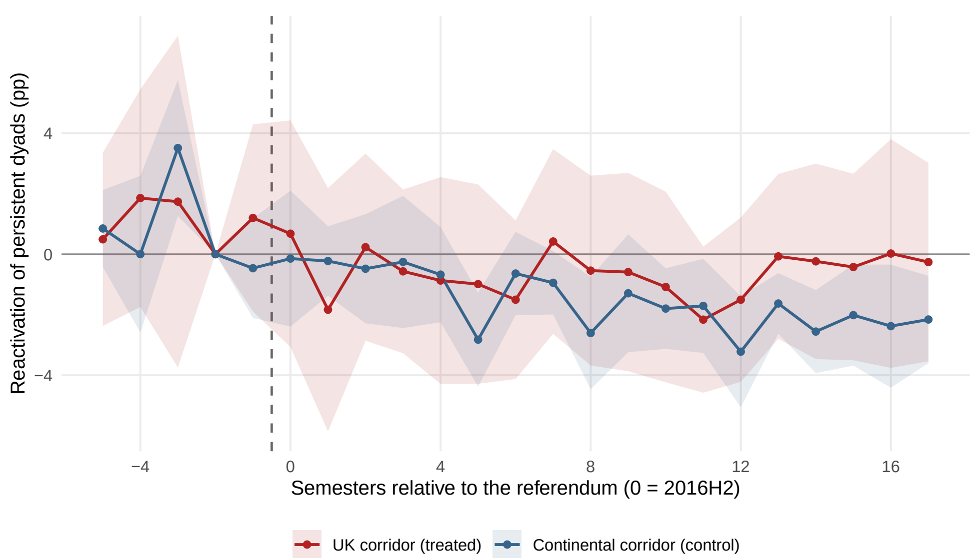

Figure 2: Reactivation of persistent dyads, UK corridor versus continental corridor. Early classification (pre-2014). The curves coincide: British ties do not die faster.

Why a naive design finds the opposite. Classifying dyads on the full pre-referendum window instead of pre-2014 yields -7.1 points, significant at all conventional levels. It also yields -5.1 points for France and -8.0 points for Germany, equally significant, with pre-trends failing in every case. A dyad observed with two or more deals just before the cut-off is, by construction, at a local peak of its activity; it reverts. The correction of Section 5 removes this entirely, and with it the divorce.

7.2 No contraction of the corridor

If ties did not break individually, did the corridor shrink in aggregate? We build a panel of country-pair corridors—the unit is an unordered pair of European countries, the outcome is asinh of the number of deals jointly syndicated by investors from both countries in a semester—and compare the eighteen UK–continental corridors against the fifty-five continental–continental corridors. Corridor and semester fixed effects absorb, respectively, the size of each corridor and the common European trend. Crucially, the outcome is defined identically for treated and control units.

The post-referendum differential is $+0.03$ log points, with $p = 0.81$ and a 95% confidence interval of $[-0.204, +0.261]$: we bound any Brexit effect on the volume of UK–continental syndication within $[-20\%, +26\%]$. Joint pre-trends are flat ($p = 0.87$) and a placebo at 23 June 2014, holding the corridor sample fixed and varying only the date, returns $+0.12$ ($p = 0.33$). Not one of the twenty-five event-study coefficients has a confidence interval excluding zero (Figure 3).

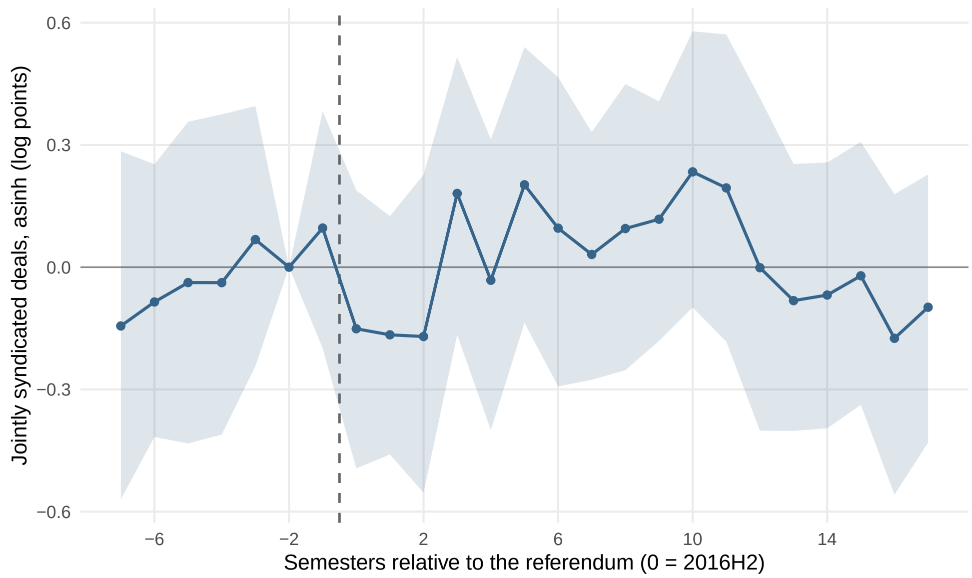

Figure 3: Volume of jointly syndicated deals: UK–continental corridors versus continental–continental corridors. Wild-cluster bootstrap-t, $B = 999$.

This confidence interval is the paper's most useful object: it is tight enough to exclude any economically meaningful change in the volume of UK–continental syndication, and we return to its precision in Appendix A.

7.3 No adaptive expansion

A natural conjecture is that investors deprived of their British channel compensate by doing more elsewhere. Comparing small, highly UK-exposed continental VCs to same-size peers never exposed to a British partner returns +6.7 percentage points ($p < 0.001$) with clean pre-trends. Taken alone, it looks like a finding.

It does not survive the battery of Section 5. Table 4 applies the identical protocol to the same thirteen corridors.

Table 4: The thirteen-corridor battery, activity. Same protocol, same grid.

Partner	Static	Pre-trends		Windows		Verdict
	(pp)	(i)	(ii)	$[0, 6]$	$[12, 17]$	
GBR	$+6.7^{***}$	0.59	0.75	+3.2 (0.18)	$+9.9^{***}$	mixed profile
USA	$+7.4^{***}$	0.17	0.26	$+5.7^{***}$	$+9.4^{***}$	**Durable response**
CHE	$+5.6^{***}$	0.41	0.57	$+4.7^{***}$	$+7.5^{*}$	**Durable response**
Non-EU OECD	$+7.7^{***}$	0.13	0.21	$+5.4^{**}$	$+10.0^{***}$	**Durable response**
NLD	$+8.7^{***}$	0.57	0.40	+3.4	$+13.7^{***}$	mixed profile
BEL	$+11.6^{**}$	0.00	0.00	+5.9	$+19.5^{***}$	pre-trends fail
CAN	+8.7	0.15	0.09	+3.3	$+15.3^{*}$	null
JPN	$+6.8^{*}$	0.89	0.83	+1.9	$+10.3^{**}$	null
SWE	+8.2	0.06	0.06	+4.0	$+14.6^{**}$	null
FRA	+1.4	0.24	0.11	+1.7	+1.5	null
DEU	−4.7	0.69	0.80	−3.5	−5.3	null

Notes. Outcome: indicator that the investor made at least one deal in the semester. Sample: small continental VCs (≤ 5 pre-2014 deals), high exposure versus never exposed. Israel and Korea are dropped for insufficient treated cells. Same protocol, same grid, same inference as Table 3.

The result fails on two independent grounds.

It is *generic*. Exposure to American partners produces +7.4 points, to Swiss partners +5.6, to a non-EU OECD aggregate +7.7—all with clean pre-trends and effects in every window. All three clear the ex-ante grid. The United Kingdom does not. Whatever this coefficient measures, it is not Brexit: it is a property of being exposed to any foreign partner.

It is *mis-timed*. In the window immediately following the referendum—H2 2016 through 2019—the UK coefficient is +3.2 points with $p = 0.18$; the 95% confidence interval is $[-1.5, +8.2]$ against a base rate of 26.0%. The effect appears only in 2020–21 (+7.8) and 2022–25 (+9.9). An effect of the referendum that is absent for three and a half years and then arrives during a global venture bubble is not an effect of the referendum.

We are careful here: this is not a precise null. The confidence interval in the post-referendum window admits effects up to +8.2 points, and with 127 treated small VCs we lack the power to exclude a genuine positive response. Our claim is not that no adaptation occurred, but that the reported adaptation is *not identified*: it cannot be distinguished from a generic network effect that appears equally for arbitrary partner countries. The same caution applies, for the same reason, to Section 7.4.

The source of the bias. The conventional control group—investors never exposed to the partner country—is not a group of comparable firms that happened to lack a British partner. Of the 757 small continental VCs never exposed to a UK investor, 548 (72%) never co-invested with *any* foreign European partner at all. They are purely domestic, unconnected firms. The comparison therefore contrasts networked investors with isolated ones, and networked investors capture a boom better, at any date. Restricting the control group to VCs that are equally networked—never exposed to the UK, but with at least one foreign continental partner—halves the estimate to +3.6 points ($p = 0.06$); restricting it instead to the isolated firms *raises* it to +7.9 points, which is the diagnostic. The coefficient is measuring the network, not the shock.

7.4 No mirror image on the British side

The universe of this paper is continental, and that choice requires a defence. The obvious reading of a severed corridor is that both ends feel it, and an earlier version of this paper reported exactly that: a British first stage of -7.1 points ($p = 0.01$), presented alongside the continental estimate as a rupture that was "sharp and symmetric on both sides." The symmetry was the more persuasive half of the claim. It is also wrong.

Nothing in the protocol of Section 5 is continental. It holds the universe of investors fixed and varies the partner corridor; setting that universe to the United Kingdom asks of a British VC exactly what Table 3 asks of a continental one—does high predetermined exposure to a given partner country predict a durable withdrawal from it? The prediction is in fact sharper on this side. Brexit changed the United Kingdom's relationship with the Union and its single market. If that is the operative channel, the EU corridors should break and the corridors outside the Union—the United States, Canada, Switzerland—should not. A directional prediction is more falsifiable than the continental side's "one corridor of thirteen," and correspondingly more dangerous to make.

Table 5: The corridor battery applied to the British panel. Identical protocol, universe set to the United Kingdom.

Partner	Static	Pre-trends		Windows		Verdict
	(pp)	(i)	(ii)	$[0, 6]$	$[12, 17]$	
Continental aggregate (the earlier draft's headline)						
Continental	$\mathbf{-6.0}^{**}$	0.04	0.03	-5.4^{*}	-5.8^{*}	**Pre-vote movement**
EU corridors: should break if Brexit is the channel						
FRA	-3.4	0.86	0.75	-1.5	-4.7	null
DEU	-4.4	0.08	0.15	-4.7	-2.5	null
NLD	-5.7	0.26	0.10	-2.2	-7.1	null
Corridors outside the Union: should not break						
USA	$\mathbf{-5.0}^{**}$	0.95	0.97	-2.3 (0.31)	-7.1^{**}	**Durable rupture**
CHE	$\mathbf{-10.9}^{***}$	0.17	0.26	-9.1^{**}	-12.2^{**}	**Durable rupture**
CAN	-1.1	0.40	0.55	-0.4	-1.4	null
Non-EU OECD	-3.9^{*}	0.73	0.84	-2.8	-5.2^{*}	null

Notes. Universe: British VCs with at least two pre-2014 deals and at least one pre-2014 partner from the corridor country; high versus low pre-2014 exposure to that country. Outcome, classification, panel, fixed effects, windows, grid and inference are those of Table 3. Clustering is on the investor rather than the investor country, which is constant on this panel; G ranges from 74 to 371, against 26 on the continental side. Sweden, Belgium, Spain, Italy, Israel, Japan and Korea are dropped by the minimum-cell rule (fewer than 30 firms in either arm). Switzerland is placed outside the Union because it belongs to neither the Union nor the single market: no Brexit channel predicts a UK–Swiss rupture. The continental aggregate follows the paper's continental definition and therefore itself includes non-EU states such as Switzerland and Norway—so it is not a clean test of Union membership either. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

The British first stage fails on two independent grounds, and they are not the same failure.

The continental aggregate is *not identified*. British VCs highly exposed to continental partners reduce their continental-partner rate by 6.0 points ($p = 0.039$), which reproduces the magnitude the earlier draft reported. But both joint pre-trend tests fail ($p = 0.04$ and $p = 0.03$): the high-exposure group was already converging towards the low one before the vote. The verdict the grid returns is mechanical—pre-vote movement, nothing identified—and it is the verdict the grid returns for Switzerland and Belgium on the continental side, where we accepted it without complaint.

The placebo corridors break *more cleanly than the treated one*. Exposure to American partners returns -5.0 points ($p = 0.032$) with pre-trends that are not merely acceptable but immaculate ($p = 0.95$ and $p = 0.97$), a negative coefficient in every window, and -7.1 points ($p = 0.018$) in 2022–25. It clears the ex-ante grid outright. With 178 treated firms it is the best-powered corridor outside the Union that we can compute, so this is not a small-sample accident: British investors who had built their deal flow around American partners withdrew from them, durably, with no pre-vote movement, and the referendum did not cause it. Switzerland clears the grid as well, at −10.9 points. Meanwhile the three EU corridors we can compute individually—France, Germany, the Netherlands—are all null. The directional prediction is not merely unmet; it is inverted.

We are careful here, as in Section 7.3. The British panel is small enough that the minimum-cell rule drops seven of the fourteen corridors, leaving three inside the Union and four outside. The French, German and Dutch nulls rest on forty to fifty treated firms and on point estimates between −3.4 and −5.8; we cannot exclude genuine European withdrawals of that magnitude, and we do not claim that British VCs demonstrably kept their continental partners. The claim is narrower and sufficient: the British first stage is not identified. Its aggregate fails both pre-trend tests, and the corridors that survive the grid with clean pre-trends are corridors Brexit cannot reach.

Why the earlier draft found symmetry. The mechanism is the one diagnosed in Section 7.1, wearing different clothes. That draft classified exposure on the full pre-referendum window and then tested pre-trends on semesters lying inside it. An investor labelled high-exposure on the basis of 2013–2016 co-investment is, over 2014–2016, high-exposure by construction: the gap between the groups is mechanically stable, and the pre-trend test it passes is a test of the classification, not of parallel trends. The early classification of Section 5 closes the window in 2013 and lets 2014–2016 evolve freely. It then shows what the earlier design had concealed—the convergence was already under way. The symmetry was an artefact of the date at which we chose to look, and it is the reason this paper's universe is continental.

8 Discussion: control-free designs manufacture mechanisms

The conventional worry about a design lacking a control group is that it is imprecise. That understates the problem. When the shock is market-wide and the aggregate is not absorbed, the trend does not inflate the standard error—it loads onto the treatment dummy. The resulting coefficient is large, correctly signed, statistically significant, and wrong. It survives the usual robustness checks, because those checks perturb the specification, not the counterfactual.

This paper contains three instances of the failure.

The dyadic divorce—a 7.1-point collapse in the survival of British ties—is regression to the mean; the identical estimate appears for France and for Germany. The adaptive expansion of exposed continental VCs is a network effect; the identical estimate appears for exposure to the United States, to Switzerland, and to a non-EU OECD aggregate, and it appears more strongly against a control group of isolated firms than against a control group of connected ones. The symmetry of the rupture—a British first stage mirroring the continental one—is a pre-vote convergence that the classification window had hidden; submitted to the battery, the British aggregate fails both pre-trend tests while withdrawal from American partners clears the grid.

None of the three is a small effect. They are -7.1 points, $+6.7$ points, and -7.1 points respectively—economically large, statistically decisive, and entirely spurious. In each case what removes them is not a better estimator but a contemporaneous counterfactual: a placebo partner country, a control group matched on network position, a placebo corridor on the treated unit's own side.

The discipline that we advocate is therefore cheap and mechanical. Fix the protocol before looking. Apply it identically to the treated corridor and to every placebo corridor for which it can be computed. Require clean pre-trends on more than one test. Require the effect to persist beyond the nearest confounding episode. And subject the treated unit to exactly the grid you apply to the placebos, accepting the verdict it returns. Applied here, this discipline destroys three of our own apparent results and leaves standing precisely one—which, as it happens, is the only one that is unique to the United Kingdom.

The symmetry case is the one we would ask the reader to weigh most heavily, because it cost us the result we liked best. Symmetry is a strong prior: if continental investors withdrew from British partners, British investors surely withdrew from continental ones, and a panel obligingly returns the coefficient. We reported it. What defeats it is not a subtler estimator but the same grid pointed at our own second panel—and the grid answers that British VCs withdrew from Americans, cleanly, durably, and for reasons a referendum on Union membership cannot supply. A design capable of certifying one side and refuting its mirror is doing the work; a design that certifies both is telling us what we expected to hear.

9 Conclusion

We summarise. First, Brexit reshuffled the composition of European venture syndicates. Continental investors who had built their deal flow around British partners durably reduced the share of their syndicates including one—by 8.2 percentage points ($p < 0.001$; 95% CI $[-13.1, -3.2]$) against a base rate of 18.7%, with clean pre-trends on two tests and a negative effect in each of three post-referendum windows through 2025. Among thirteen partner corridors subjected to an identical, pre-specified protocol, the United Kingdom is the only one where this occurs.

Second, that compositional shift severed nothing. Established British ties were no more likely to dissolve than comparable continental ones ($+0.3$ pp, $p = 0.77$; we exclude any reduction exceeding a third of the baseline survival rate), and the aggregate volume of UK–continental syndication tracked the rest of Europe within a band of $[-20\%, +26\%]$. Persistent relationships decay everywhere, at the same rate, whether the partner sits in London or in Milan.

Third, the apparent adaptation of exposed investors is not identified. It appears with equal or greater strength for exposure to American, Swiss, and non-EU OECD partners; it is absent from the window that follows the referendum (+3.2 pp, $p = 0.18$) and materialises only during the venture boom of 2020–2021; and its conventional control group turns out to be 72% composed of investors who had never syndicated abroad at all. Every remaining outcome we can identify—deal count, stage, target geography, follow-on financing, exits—is null.

Fourth, the rupture has no British counterpart. The same battery, run on a British universe where the directional prediction is sharper—Brexit altered the United Kingdom's relationship with the Union and with nothing else—returns a continental aggregate that reproduces in magnitude (−6.0 pp, $p = 0.039$) but fails both pre-trend tests ($p = 0.04$, $p = 0.03$), and clean, durable withdrawals from precisely the corridors the referendum cannot reach: the United States (−5.0 pp, pre-trends $p = 0.95$) and Switzerland (−10.9 pp). France, Germany and the Netherlands are null. The symmetry we once reported was an artefact of classifying exposure on the window we then tested for parallel trends.

The relationships survived. What changed was their weight, and it changed on one side only.

We close on the methodological point, because it generalises beyond this setting. Brexit is the archetype of a shock in which no agent is untreated, and such shocks are common in finance: a regulatory change, a currency regime break, a pandemic. In all of them the temptation is to compare before with after and to name the difference a mechanism. The econometric literature has taught us to worry about heterogeneous effects and staggered adoption; it has less to say about the case where there is simply nothing to compare against, which is the harder case and, we suspect, the more common one. The three mechanisms we examined here—a dyadic divorce, an expansionary response to lost partners, and a symmetric withdrawal on the British side—were all large, all significant, and all spurious. They failed identically, and the failure has a single signature: the coefficient was the aggregate trend, wearing the treatment's clothes.

The discipline we advocate destroyed three of our own apparent results and left standing only the one that no placebo corridor could reproduce. We regard that as the paper's principal demonstration, and not as a caveat to it.

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A Precision: what the nulls exclude

A null result is informative only if it is precise. Table 6 reports, for each headline estimate, the wild-cluster bootstrap confidence interval obtained by inverting the bootstrap distribution, the base rate of the outcome in the control group before the referendum, and the minimum detectable effect at 80% power.

Table 6: Precision of the headline estimates.

		Estimate	95% CI	Base rate	MDE
Effect	First stage: UK-partner rate (pp)	−8.20	$[-13.11, -3.15]$	18.7%	6.64
Null	Dyad survival, triple diff. (pp)	+0.30	$[-1.84, +2.51]$	5.6%	2.62
Null	Corridor volume (log pts)	+0.03	$[-0.20, +0.26]$	7.5 deals	0.32
Null	Activity, window $[0, 6]$ (pp)	+3.18	$[-1.47, +8.16]$	26.0%	6.17
Not identified	British mirror: first stage, UK panel (pp)	−6.01	$[-11.52, -0.33]$	22.5%	8.51

Notes. Confidence intervals from inversion of the wild-cluster bootstrap-t distribution (Rademacher, unrestricted, equal-tailed, $B = 999$). Base rate: mean of the outcome among control units before the referendum. MDE: minimum detectable effect at 80% power and 5% size. The last row is not a null—its interval excludes zero—and is reported here for the diagnostic in the text.

Two of the three nulls are informative and one is not, and we distinguish them. The British estimate is not a null at all, and we distinguish it too.

The corridor null is decisive. Its interval bounds any Brexit effect on the volume of UK–continental syndication within $[-20\%, +26\%]$. Large increases in cross-border syndication of the kind that have been read off the post-2016 data—rises on the order of 80% in the cross-border share—lie far outside it: once a contemporaneous control corridor is added, no effect of that magnitude survives.

The dyad null is informative. We exclude any reduction in the semi-annual survival of British ties exceeding 1.8 percentage points, one third of the 5.6% baseline reactivation rate. A "divorce" of any economically meaningful size is ruled out.

The activity null is *not* precise. Its interval admits effects up to +8.2 points against a base rate of 26%, and with 127 treated small investors we lack the power to exclude a genuine adaptive response. Our claim about activity therefore rests on the falsification battery of Section 7.3, not on precision: the estimate is indistinguishable from what the same design returns when the partner country is the United States or Switzerland, and it is absent from the window that follows the referendum. We report it as unidentified, not as zero.

The British mirror is the opposite case, and it is worth being explicit about why it sits in this table. Its interval, $[-11.5, -0.3]$, excludes zero: read on its own, it is a significant 6-point withdrawal, and an earlier version of this paper read it exactly so. Precision is not what disqualifies it—Section 7.4 does, on pre-trends and on placebo corridors. But the table adds an independent warning that we had missed. The minimum effect this panel can detect with 80% power is 8.5 points, larger than the 6.0 points it reports. A design that returns a significant coefficient below its own minimum detectable effect is not delivering a precise finding; it is drawing from the tail in which significant estimates are, mechanically, the overstated ones. The British panel has fewer investors than the continental one and a higher base rate, and it is underpowered against effects of the size it appears to find. That the same panel returns a cleanly identified withdrawal from *American* partners is the sharper objection, but this one is available to any reader with the table.

B The remaining outcomes

We list every outcome we examined, so that the selection of reported results can be verified rather than trusted. All are estimated on the same sample as the first stage—continental VCs, all UK-exposed, high versus low—with early classification, investor and semester fixed effects, a small × post control, and wild-cluster bootstrap inference.

Table 7: All secondary outcomes.

Outcome	Estimate	95% CI	p	Pre-trends
Deal count (asinh)	−0.03	[−0.15, +0.08]	0.52	0.53
Early-stage share (pp)	−0.83	[−8.30, +8.08]	0.81	0.20
Continental-target share (pp)	+2.47	[−8.09, +13.38]	0.55	0.24
Follow-on round within 4 years (pp)	+0.16	[−8.96, +8.96]	0.96	0.83
Exit, M&A or IPO, within 4 years (pp)	+4.72	[−3.00, +13.35]	0.16	0.13
Not identified—fails the pre-trend test:				
Mean round size (asinh)	−0.09	[−0.23, +0.05]	0.21	0.00

Notes. Pre-trends: joint Wald test on $\tau \in \{-5, -4, -3\}$. Mean round size fails it and we therefore report no estimate for it: the groups were already diverging before the referendum, and nothing is identified. We do not interpret its coefficient, and it should not be read as a null.

Every outcome we can identify is null. Exposed investors did not do more or fewer deals, did not move to earlier or later stages, did not redirect their deal flow toward continental targets, and their portfolio companies were neither less likely to raise again nor less likely to exit. One outcome—the average size of the rounds they participate in—fails the pre-trend test outright, and we decline to interpret it rather than report it as a null. This matters: a paper whose contribution is that control-free designs manufacture mechanisms cannot afford to report an unidentified coefficient as evidence of anything, in either direction.